

PITP ASSIGNMENT

STEGHIDE TOOL

MEMBERS:

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Report: Hiding Data in Audio Extracted from Video using Steghide

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1. Introduction

Objective: Demonstrate how to hide a secret file inside audio extracted from a video using FFmpeg and Steghide.
Tools Used:

- **Kali Linux**
- **FFmpeg**
- **Steghide**
- **Open-source video (Big Buck Bunny, CC license)**

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2. Video Download

Source: Big Buck Bunny (Creative Commons licensed demo video).
Command Used:

wget http://commondatastorage.googleapis.com/gtv-videos-bucket/sample/BigBuckBunny.mp4

Result: The video file BigBuckBunny.mp4 was successfully downloaded and stored locally.

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3. Audio Extraction

Command Used:

ffmpeg -i BigBuckBunny.mp4 -vn -acodec pcm_s16le -ar 44100 -ac 2 output.wav

Explanation:

- **-vn removes video stream.**
- **pcm_s16le ensures uncompressed audio (lossless).**
- **-ar 44100 sets sample rate to 44.1 kHz.**

- **-ac 2 sets stereo channels.**

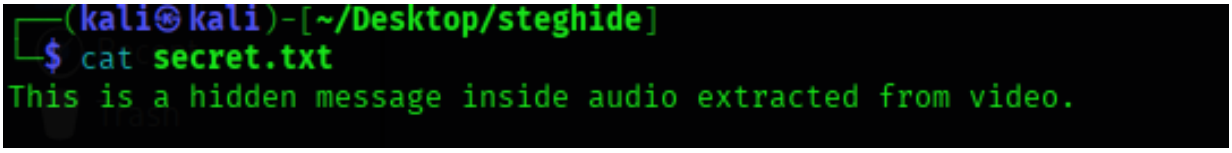
Result: Audio extracted into output.wav.

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4. Secret File Creation

File Name: secret.txt
Contents:

This is a hidden message inside audio extracted from video.



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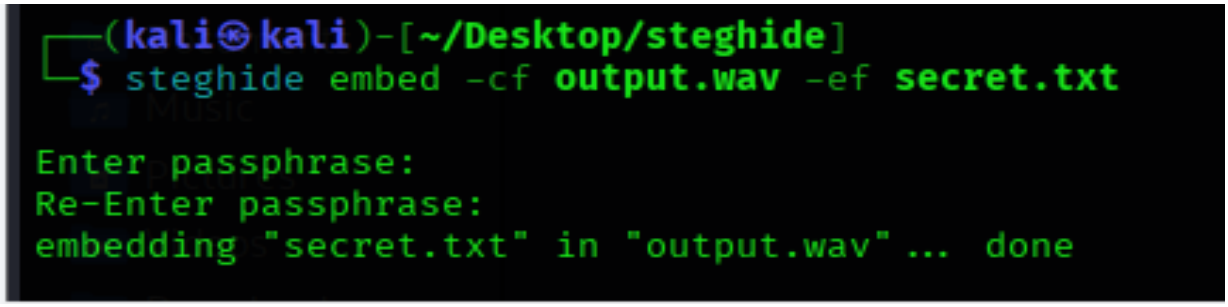
5. Embedding Data with Steghide

Command Used:

steghide embed -cf output.wav -ef secret.txt

Explanation:

- **-cf specifies cover file (output.wav).**
- **-ef specifies file to embed (secret.txt).**
- **Steghide prompts for a passphrase and embeds the file securely.**



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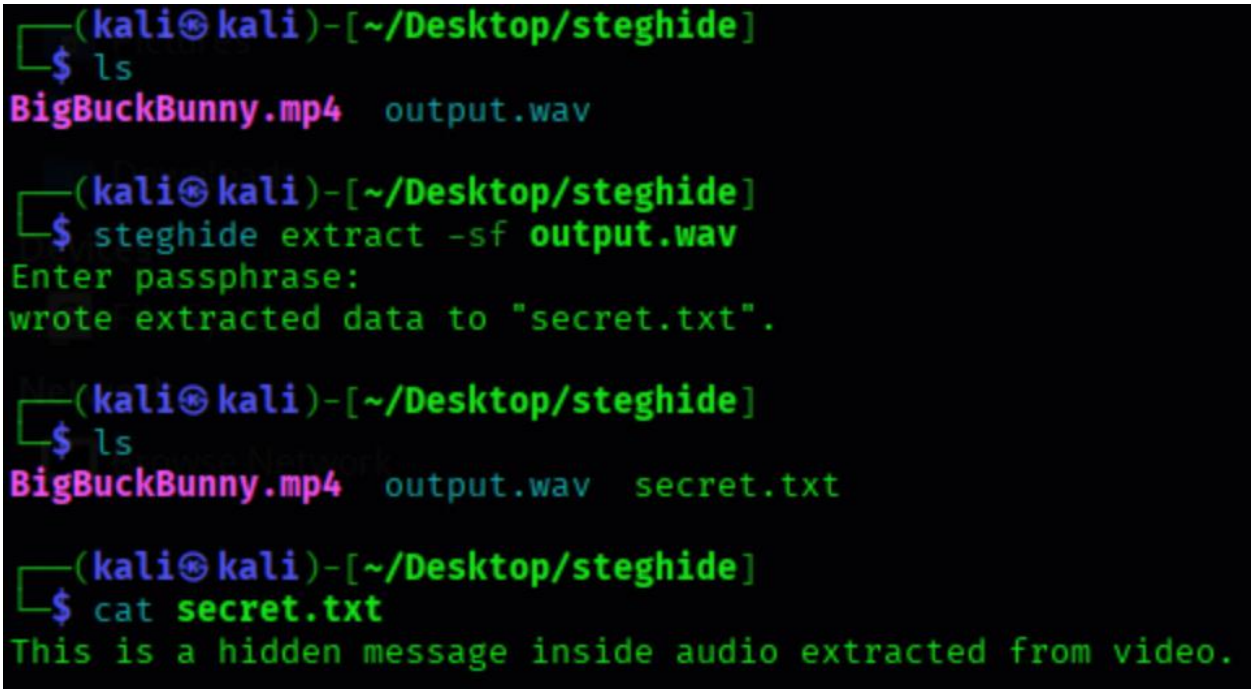
6. Extraction Process

Command Used:

steghide extract -sf output.wav

Recovered File:

This is a hidden message inside audio extracted from video.

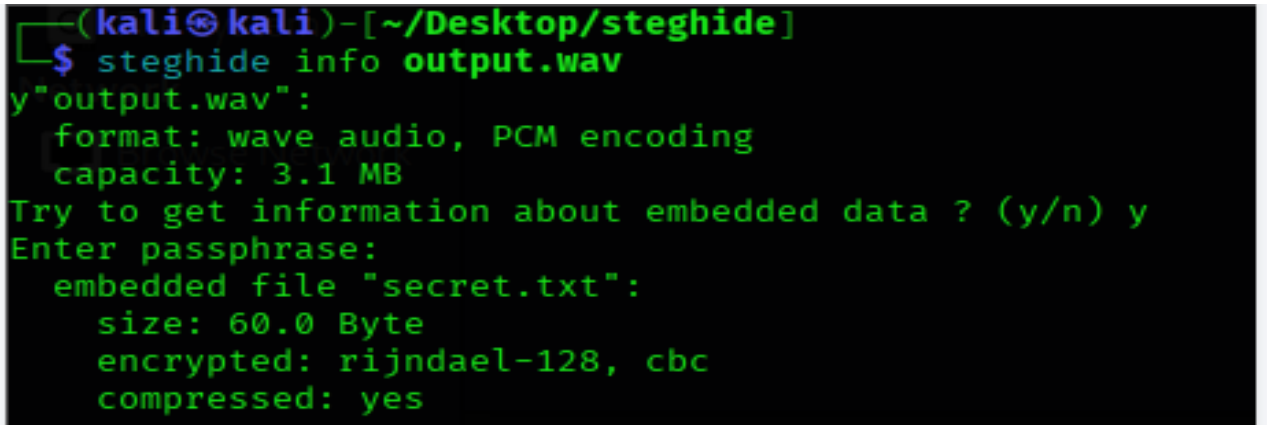


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7. Verification

Using steghide info output.wav confirmed:

- **Embedded file: secret.txt**
- **Size: 60 Bytes**
- **Encrypted: Rijndael-128, CBC mode**
- **Compressed: Yes**



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8. Observations

- **Audio file size increased slightly after embedding.**
- **Playback of output.wav remained unchanged — hidden data is imperceptible.**
- **Extraction successfully restored the original secret file.**

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9. Conclusion

- **Steghide effectively hides and retrieves data in audio extracted from video.**
- **FFmpeg provides a reliable way to separate audio streams for steganography.**
- **The experiment confirmed that the hidden message was securely embedded and extracted without altering audio quality.**